

Mental Health and Psychosocial Outcomes Among Individuals With Mayer-Rokitansky-Küster-Hauser (MRKH) Syndrome: A Scoping Review

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Abstract

Background: Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome is a rare congenital condition (1 in 4,500–5,000 female births) characterized by uterovaginal agenesis in individuals with a 46,XX karyotype and functional ovaries. Diagnosed during adolescence, MRKH is associated with significant psychosocial burden. Mental health outcomes and coping mechanisms remain insufficiently synthesized, limiting integration into clinical care and public health planning.

Methods: Following Arksey and O'Malley's framework and PRISMA-ScR guidelines, we searched PubMed/MEDLINE, Scopus, PsycINFO, and CINAHL (January 2019–March 2026). Eligible studies included primary quantitative, qualitative, and mixed-methods research on psychological outcomes and/or coping mechanisms in MRKH populations.

Results: 34 studies met inclusion criteria. Depression and anxiety were most frequently reported (n = 26), followed by reduced QoL and body image concerns (n = 22), psychosexual challenges (n = 19), and psychological distress (n = 17). Coping mechanisms were documented in 21 studies; healthcare gaps in 18.

Conclusions: MRKH-related psychosocial burden is substantial yet under-integrated into care models. Multidisciplinary, mental health-inclusive care and greater research investment are urgently needed.

Keywords: MRKH syndrome; mental health; coping; psychosocial outcomes; quality of life; reproductive health

Introduction and Research Question

MRKH syndrome is a rare Müllerian aplasia resulting in congenital absence of the uterus and upper two-thirds of the vagina in chromosomally female individuals (46,XX) with functional ovaries. It affects approximately 1 in 4,500–5,000 female births^{1,2} and is typically diagnosed during adolescence, often during evaluation for primary amenorrhea¹—a critical period for identity formation and social development.

The diagnosis carries profound implications for fertility, psychosexual development, and self-image. Professional guidance identifies psychosocial counseling as a key component of care,¹ yet psychosocial outcomes remain underexplored and insufficiently integrated into clinical and public health frameworks.

Research Question

What mental health outcomes and coping mechanisms are reported among individuals with MRKH syndrome, and what healthcare system gaps exist?

Learning Objectives

- Identify key mental health outcomes—including depression, anxiety, reduced quality of life, and psychosexual challenges—reported among individuals with MRKH syndrome.
- Describe coping mechanisms and psychosocial responses, including adaptive and maladaptive strategies, documented in MRKH populations.

Methods

Category	Description
Study Design	Scoping review following Arksey & O'Malley's five-stage framework; ³ reported per PRISMA-ScR ⁴
Databases	PubMed/MEDLINE, Scopus, PsycINFO, and CINAHL
Years Searched	January 2019 – March 2026
Inclusion Criteria	Primary quantitative, qualitative, and mixed-methods studies reporting psychological outcomes and/or coping mechanisms in MRKH
Exclusion Criteria	Studies focused solely on anatomical, surgical, or fertility outcomes
Screening	516 records identified; 155 duplicates removed; 361 screened; 327 excluded; 34 included
Synthesis	Descriptive frequency counts by outcome domain and thematic grouping of coping mechanisms and healthcare gaps

MRKH Syndrome at a Glance

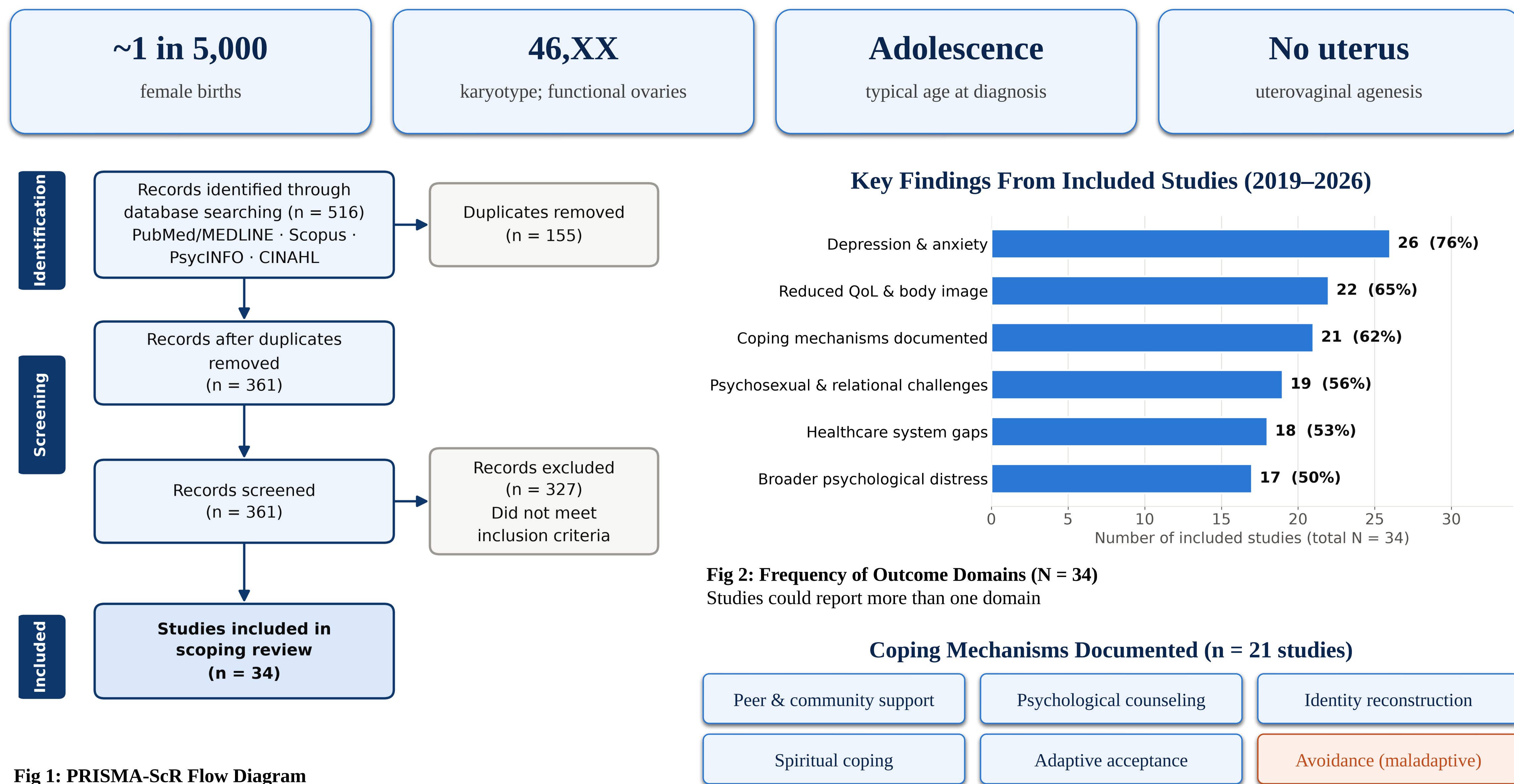


Fig 1: PRISMA-ScR Flow Diagram
(Adapted from Tricco et al., 2018)

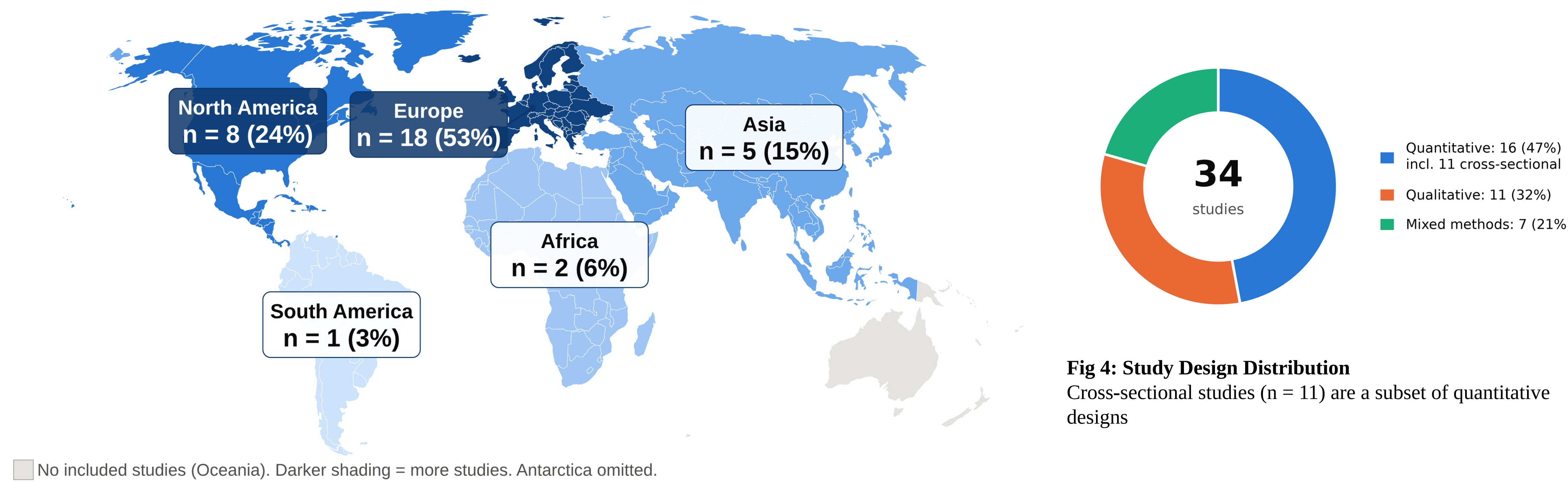


Fig 3: Geographic Distribution of Included Studies
26 of 34 studies (76%) from Europe or North America; 3 from Africa or South America

Fig 4: Study Design Distribution
Cross-sectional studies (n = 11) are a subset of quantitative designs

Results, Discussion and Conclusion

Results

Of 516 records identified, 361 remained after deduplication and 34 studies met inclusion criteria: 16 quantitative (11 cross-sectional), 11 qualitative, and 7 mixed-methods. Depression and anxiety were reported in 26 studies (76%), reduced QoL and body image in 22 (65%), psychosexual and relational challenges in 19 (56%), and broader distress in 17 (50%). Healthcare system gaps were identified in 18 studies (53%): delayed diagnosis without psychosocial support, limited multidisciplinary care, mental health insufficiently integrated into care models, and no standardized screening at diagnosis.

Discussion

These findings are consistent with earlier evidence of elevated distress, anxiety, and poorer mental health-related quality of life in MRKH^{5–7} and with a prior systematic review.⁸ Coping strategies were documented in most studies but remain unsupported by systematic clinical pathways, and 76% of studies came from Europe or North America, leaving low-resource settings underrepresented.

Conclusion

MRKH-related psychosocial burden is substantial yet under-integrated into care models. Multidisciplinary, mental health-inclusive care and greater research investment are urgently needed, particularly in underrepresented and low-resource settings.

Limitations

- Review limited to 2019–2026, potentially excluding earlier foundational research
- Heterogeneous designs and outcome measures limit direct comparisons
- Most studies from high-income countries, limiting global generalizability
- Some studies relied on self-reported mental health outcomes

Public Health Implications

- Integrate mental health screening into MRKH diagnostic pathways
- Develop multidisciplinary care protocols including psychology and social work
- Expand research investment in LMIC and underrepresented settings
- Build peer support infrastructure within reproductive health services
- Train providers in psychosocially informed MRKH care

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